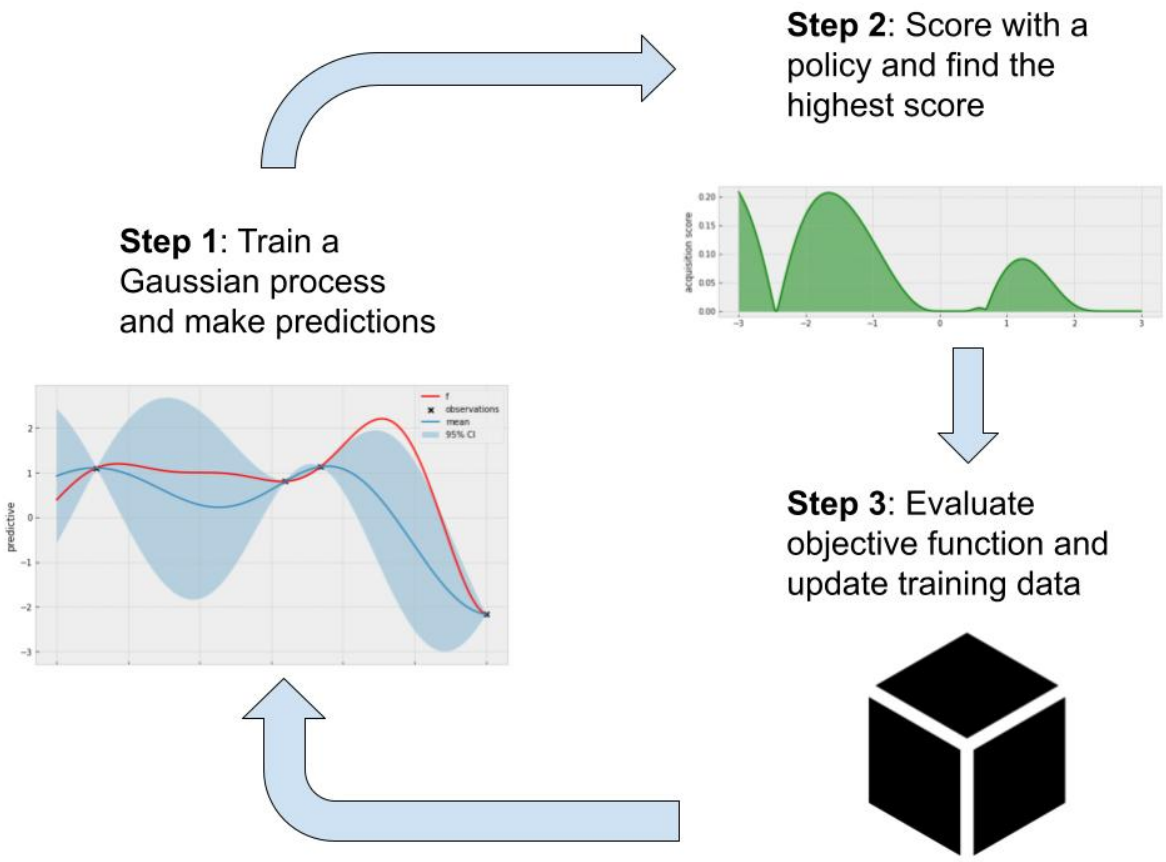


Figure 1.6 The Bayesian optimization loop, which combines a GP for modeling and a policy for decision making. This complete workflow may now be used to optimize black-box functions.

Unlike a supervised learning task in which we just fit a predictive model on a training data set and make predictions on a test set (which only encapsulates steps 1 and 2 above), a Bayesian optimization workflow is what's typically called *active learning*, a subfield in machine learning where we get to decide which data points our model learns from, and that decision-making process is in turn informed by the model itself.

As we have said, the GP and the policy are the two main components of this Bayesian optimization procedure. If the GP does not model the objective well, then we will not be able to do a good job informing the policy of the information contained in the training data. On the other hand, if the policy is not good at assigning high scores to "good" points and low scores to "bad" points (where "good" here means helpful in helping locate the global optimum), then our subsequent decisions will be misguided and most likely achieve bad results.

In other words, without a good predictive model such as a GP, we won't be able to make good predictions with calibrated uncertainty. Without a policy, we can make good predictions but we won't make good *decisions*.

An example that we will consider multiple times throughout this book is weather forecasting. Imagine a scenario where you are trying to decide whether to bring an umbrella with you before

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